

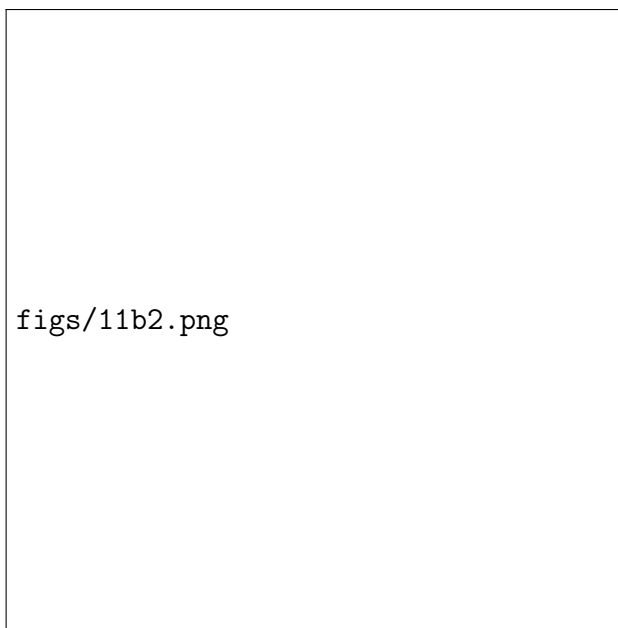
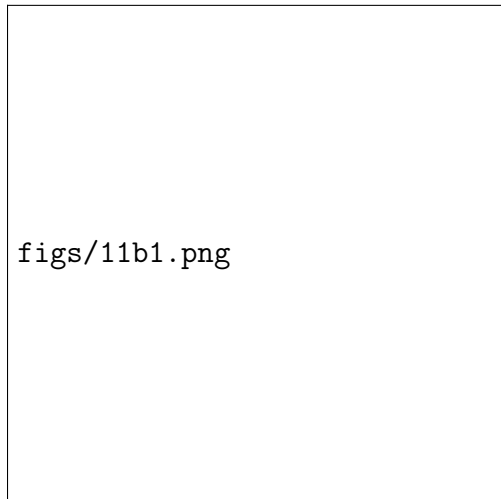
Notes for Externalities and Public Goods

Externalities

Consider an interior solution. For negative externalities, $\phi'_2 < 0$.

- Competitive outcome: $\phi'_1(h^*) = 0$.
- Social planner sets $\phi'_1(h^\circ) = -\phi'_2(h^\circ) > 0$.
- $\phi'_1(h^\circ) > \phi'_1(h^*)$, so $h^* > h^\circ$ (since ϕ'_1 is strictly decreasing).

Negative externality:



For positive externalities, $\phi'_2 > 0$.

- Competitive outcome: $\phi'_1(h^*) = 0$.
- Social planner sets $\phi'_1(h^\circ) = -\phi'_2(h^\circ) < 0$.
- $\phi'_1(h^\circ) < \phi'_1(h^*)$, so $h^* < h^\circ$ (since ϕ'_1 is strictly decreasing).

Coase Theorem

Distribution of surplus for each combination. The total surplus is the same in each.

R	B	Problem	v_1	v_2
1	1	$\max_{h \geq 0, T} \phi_1(h) + T$ s.t. $\phi_2(h) - T \geq \phi_2(h^*)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_2(h^*)$	$\phi_2(h^*)$
1	2	$\max_{h \geq 0, T} \phi_2(h) - T$ s.t. $\phi_1(h) + T \geq \phi_1(h^*)$	$\phi_1(h^*)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_1(h^*)$
2	1	$\max_{h \geq 0, T} \phi_1(h) - T$ s.t. $\phi_2(h) + T \geq \phi_2(0)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_2(0)$	$\phi_2(0)$
2	2	$\max_{h \geq 0, T} \phi_2(h) + T$ s.t. $\phi_1(h) - T \geq \phi_1(0)$	$\phi_1(0)$	$\phi_1(h^\circ) + \phi_2(h^\circ) - \phi_1(0)$

figs/11b3.png

- a is where 2 has the property rights and $(\phi_1(0), \phi_2(0))$
 - d is where 1 bargains and f is where 2 bargains.
- b is where 1 has the property rights and $(\phi_1(h^*), \phi_2(h^*))$
 - c is where 1 bargains and e is where 2 bargains.